

GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING SCIENCES AND TECHNOLOGY

ANNUAL ACADEMIC RECORD

(FILLED BY FACULTY MEMBERS)

Year: 2025

NAME HAMZA ABBAS

FACULTY: FME

1. TEACHING:

Semesters - Fall and Spring

Year/Sem.	No. of Sects.	Course No.	Credit/ Lab. Hrs.*	Course Title	Enrolment	Function	Student Evaluation	
							No. of Std. respond.	Score
Fall 2025	1	ME 439	3/0	Heating, Ventilation and Air-Conditioning	10	T	10	65.82 %
Fall 2025	2	ME 346	1/3	Thermo Fluid Lab 1	68	C	64	71.87 %
Spring 2025	2	ME 212	6/0	Dynamics	72	T	55	74.00 %

*Contact hours 12

Senior Project	0	Function: T: Teaching C: Coordinating
Summer Training:	0	

SHORT COURSES AND OTHER TEACHING ACTIVITIES

Year /Sem	Hours Taught.	Short course Title and duration	No of Participants	Function	Course Fee/ Free	Participants Evaluation
Fall 2025	1	HAP Training for Students of ME439	10	Teaching	Free	N/A

PART 1: STUDENT EVALUATION:

(Source: Student evaluation in courses, graduating student evaluation, evaluation by participants of short courses, and other sources). Please state your comments, if any, on student evaluation:

The qualitative feedback for ME439 Fall 2025 indicates strong student engagement and successful learning outcomes, specifically regarding the HVAC plant visit and air handling units—the quantitative data shows

variance that does not match the written praise. To ensure more reliable data in future cycles, it may be beneficial to emphasize the importance of evaluation accuracy to the students to ensure their ratings better reflect their written feedback.

Part 2 COURSE MATERIAL COURSE FILE:

(Activities: Organization of course file, development of course material and new courses, adherence to course syllabus, development and use of Innovative methods in teaching, Instructional laboratories, and other activities related to course material development). Please list your activities with regard to course material and comment on them:

- Changed ME439 course title from “Refrigeration and Air-Conditioning” to “Heating, Ventilation and Air-Conditioning” and introduced few new topics like Refrigerant Selection.
- Taken Class/Suprise Quizzes in ME439 and ME212. This method promotes active engagement, and student-centered learning.

PART 3: OTHER TEACHING ACTIVITIES:

(Advising, supervising senior projects, summer training and students internships, initiation and participation in curriculum development, course co-ordination, and participation in student activities, etc.) Please list and comment on your other teaching activities.

- **Social Media Coordinator FME** – Managing FME Facebook and LinkedIn Page.
- **Batch Advisor (Batch 34) (Non-Lead).**
- **Lab Incharge** – Fluid Mechanics Lab, FME
- FME and Management Elective courses replacement with Coursera/Edx courses.
- Helped in writing criterion 0 of the **SAR**.

II RESEARCH AND SCHOLARSHIP

Report Activities for the past academic year only. However, please note that your research profile for the past Two years will be evaluated:

A. RESEARCH ACTIVITIES:

1. Funded Research Project:

Status

Role

Time:Spent

C: Completed
IP: In Progress
T: Terminated

PI: Principal Investigator
CI : Co-Investigator

Percentage of Time as
Approved

Project Title	Funding Agency (Amount in Rs.)	Start Date	End Date	Status	Role	Time Spent
Industrial Water Heating System with Grid-Tied PV Integration	Pakistan Tobacco Company	04/2025	To Present	IP	CI	10 %

2. Independent Research

Research Title	Start Date	End Date	Hrs/Wk.	Expected Output and Date Journal/Conference publications, engineering designs, software, technical report etc.
Cooling and Heating load Calculation	Dec, 2025	Feb, 2026	1	1 CP

3. Thesis and Dissertation Supervised

Degree

Role

Msc or Ph.D.

S: Supervisor

CM: Thesis/Dissertation Committee Member

MS and PhD Research

Student Name	Dept.	Deg.	Thesis Title	Start Date	End Date	Role

B: PUBLICATIONS FOR THE LAST TWO YEAR

Please list out by separating into the following:

Class:

JP: Journal Publication

TRP: Technical Reports & Proposals

CP: Conference Publication

PPW: Published Professional Work e.g. Engg. Design

OP: Other Publication

PTJ: Publication in Trade journal

Status:

A: Accepted/Published

S: Submitted

UP: Under Preparation

CLASS	PUBLICATIONS (FOR THE PAST TWO YEARS ONLY)	STATUS
JP	Hamza Abbas , Ali Turab Jafry*, Mubashar Ali, Taqi Ahmad Cheema, “Impact of paper thickness, contact area, and fluid volume on flow dynamics in paper-based microfluidics” (Submitted in: Langmuir ACS).	S
JP	Mubashar Ali, Ali Turab Jafry*, Hamza Abbas , Minki Lee, Muhammad Salman Abbasi, Syed Farhad Shah, Huma Ajab, Muhammad Kamran, Ali Haider Jafry. “Revealing the Mechanics of Highly Resistive Fluid Flow in a Leaf Architecture using Paper Microfluidics”. (Submitted in: Journal of Fluid Mechanics)	S
CP	Ullah, H., Ali, M., Naeem, N., Iqbal, F., Abbas, H. , Jafry, A. T., & Ajab, H. (2024). Droplet Actuation Enhancement through Voltage Control and Hydrophobic Coating Selection. In MATEC Web of Conferences (Vol. 398, p. 01020). EDP Sciences	A

III COMMITTEES AND OTHER ACTIVITIES

(Contribution to meet the Objectives of the Institute)

A: COMMITTEE WORK

FORMED BY: POSITION TYPE OF COMMITTEE

F: Faculty M: Member S: Standing
 I: Institute C: Chairman a: Ad-Hoc
 G: Other (state)

Name of Committee	Semester	Academic Year	Formed By	Position	Type	App. No of Attended meetings
Faculty Study Board	Fall 2025	2025	F	M	S	1
GIKI LMS Support	Spring 2025, Fall 2025	2025	I	M	S	2
OGDCL Workshop	Summer 2025	2025	F	M	a	1

Please describe your major activities and contribution to each committee (Use additional sheets, if necessary).
 Supervision and mentoring of students, co-curricular and extra-curricular activities and other programs,
 Student Advisor and Student Society Advisor, any other contribution to meet the Objectives of the Institute.

- **Faculty Study Board** – Suggested few courses title changes & noting Minutes of the Meeting.
- **GIKI LMS Support** – Working as a LMS Coordinator FME
- **OGDCL Workshop** – Worked as a lead organizer

IV. KNOWLEDGE EXCHANGE**A TECHNICAL SEMINARS AND PRESENTATIONS:**

Title	Date	Place
Pipeline Integrity (Engr. Adeel Raza)	12 th Dec, 2025	LH3-FME, GIK Institute
Faculty Training Session – Summer 2025 Topics: 1) Best Practices in Research Proposal Writing 2) Learning Through Problems	28 th August 2025 – 29 th August 2025	New Academic Block, GIK Institute

B PROFESSIONAL CONSULTING:

Organization	Nature of Work	Hrs./Wk	Start Date	End Date

C OTHER PROFESSIONAL ACTIVITIES:

(List patents received, works reviewed, funding received, international and national linkages, new initiatives – lab development, curriculum review new program)

Patent in Submission: “An enclosed fan shaped flexible time-temperature indicator” (Local)

V: CPD AND OTHER ACTIVITIES

Please list any other activity you have undertaken during the Year

Date	Activity
24/07/2025	Graduate Admission Test Invigilation (Fall 2025)
02/12/2025	Led Industrial Trip to KSB Pumps
20/04/2025	Assisted in hosting the MTME-2025 Conference
27/08/2025	Attended Seminars for Faculty Position
01/2025	Graduate Admission Test Invigilation (Spring 2025)

The information given in this form is correct, accurate and complete.

Name **Hamza Abbas**

Signature/and Date:  15/01/26

Faculty: FME